UNIT 3

Energy Resources, Electricity Bill & Safety Measures

Topics: Conventional & non-conventional resources • Hydel, nuclear, solar & wind generation • Appliance power ratings • Unit (kWh) • Two-part tariff • Bill calculation • Fuse & MCB • Electric shock • Earthing • Safety precautions

START HERE
Big picture: Electricity must be **generated** from some energy resource, **paid for** by consumers (the electricity bill) and **used safely**. This unit covers all three.

1 ENERGY RESOURCES

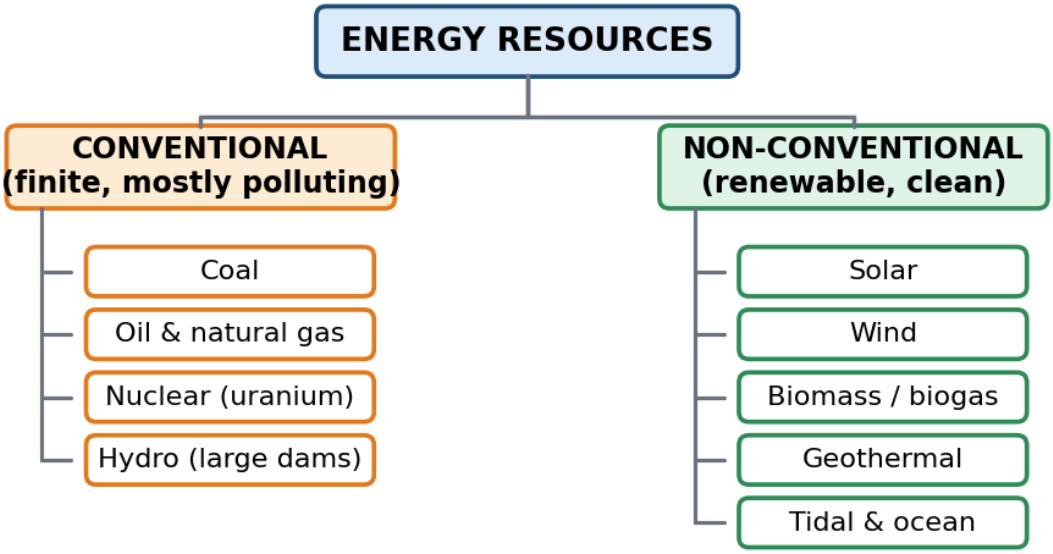


Fig 1 Classification of energy resources

Feature	Conventional	Non-conventional
Availability	Limited (fossil fuels, uranium); being depleted	Unlimited; renewable
Pollution	High (CO_2, SO_2, ash, nuclear waste)	Very low / nil
Cost	High running cost, low set-up cost (thermal)	High set-up cost, almost free fuel
Power level	Large, steady, reliable	Smaller, depends on weather/location
Examples	Coal, oil, gas, nuclear, large hydro	Solar, wind, biomass, geothermal, tidal

1.1 Hydel (Hydroelectric) Power Generation

DEFINITION

Working idea: potential energy of water stored at a height → kinetic energy as it falls through the **penstock** → mechanical energy in the **turbine** → electrical energy in the **generator**.

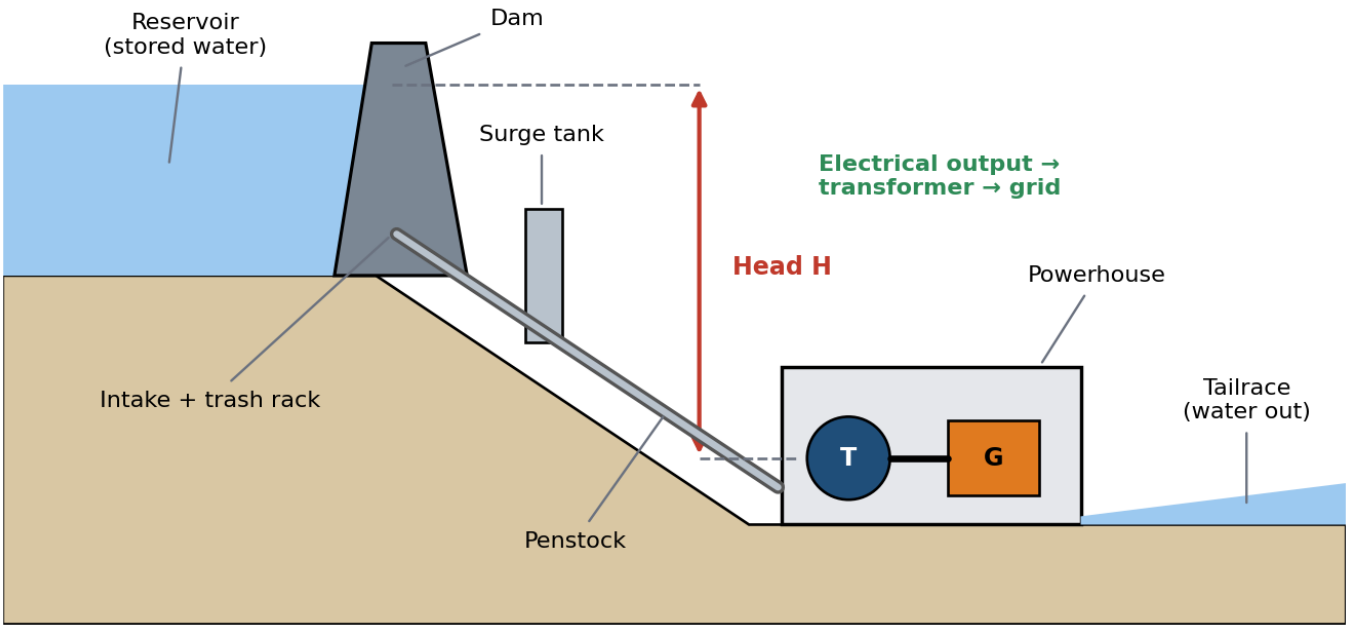


Fig 2 Layout of a hydroelectric power plant

$P = 9.81 \times Q \times H \times \eta$ kW
(Q = flow in m³/s, H = head in m, η = overall efficiency)

Component	Function
Dam + reservoir	Stores water and creates the head H
Intake / trash rack	Admits water and filters out debris
Penstock	Steel pipe carrying water to the turbine
Surge tank	Absorbs pressure surges (water hammer) when load changes
Turbine	**Pelton** (high head), **Francis** (medium head), **Kaplan** (low head)
Tailrace	Channel returning used water to the river

- **Merits:** no fuel cost, clean, long life, quick start, also gives irrigation and flood control.
- **Demerits:** very high initial cost, depends on rainfall, needs large land area, displaces people/ecology.

1.2 Nuclear Power Generation

DEFINITION

Working idea: In the reactor, neutrons split enriched **uranium-235** atoms (**nuclear fission**); each fission releases heat and more neutrons, sustaining a **chain reaction**. The heat makes steam, which drives a turbine-generator, just like a thermal plant but with the reactor replacing the boiler.

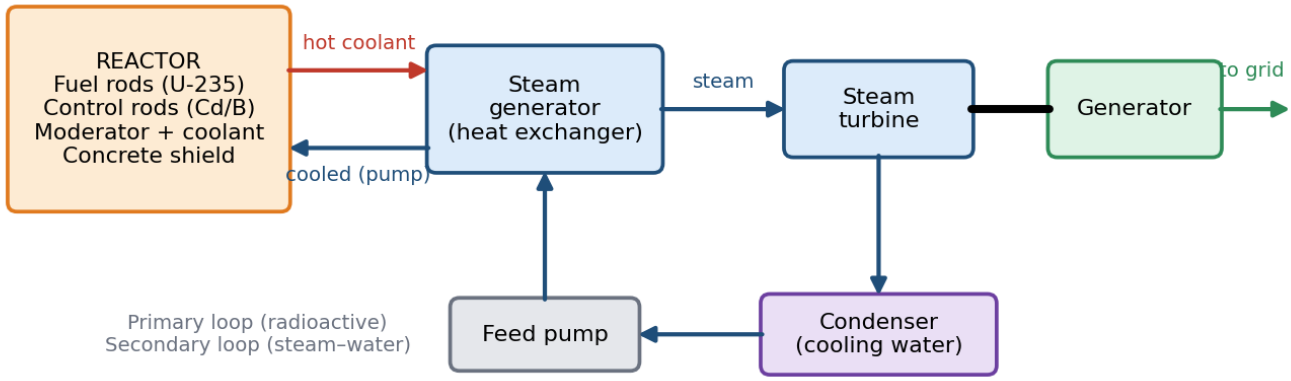


Fig 3 Block diagram of a nuclear power plant

Part	Role
Fuel rods	Enriched uranium (U-235) pellets; source of fission heat
Moderator	Slows neutrons so they cause fission (heavy water, graphite, ordinary water)
Control rods	Cadmium / boron absorb neutrons; inserted to slow or stop the reaction
Coolant	Water / heavy water / CO_2 carries heat to the steam generator
Shielding	Thick concrete + steel to stop radiation

- **Merits:** huge energy from a tiny amount of fuel, no CO_2 during operation, steady base-load power.
- **Demerits:** radioactive waste disposal, very high capital cost, risk of accidents, strict safety needs.

1.3 Solar Power Generation

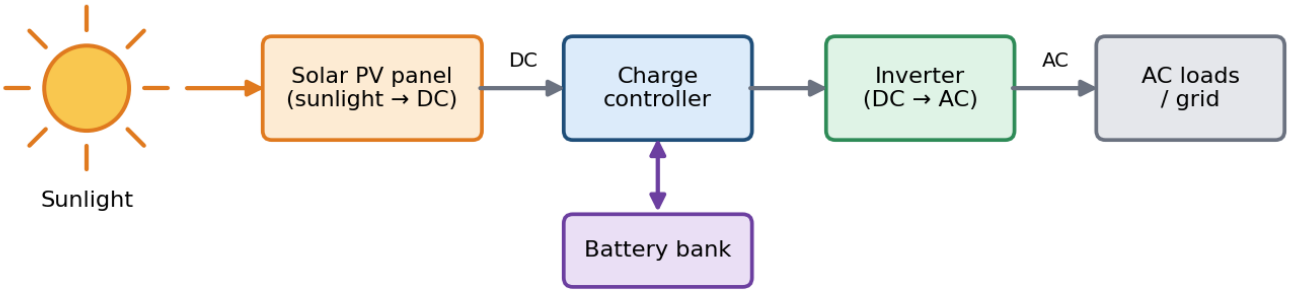


Fig 4 Stand-alone solar photovoltaic (PV) system

A **solar cell** (silicon p-n junction) converts sunlight directly into **DC** by the **photovoltaic effect**. Many cells form a module; modules form an array. The **charge controller** protects the battery, the **battery** stores energy for night use and the **inverter** converts DC to AC. In *grid-connected* systems the battery is omitted and extra power is fed to the grid.

- **Merits:** free, clean, silent, low maintenance, good for remote areas.

- **Demerits:** works only in daylight, high initial cost, needs large area, output varies with weather.

1.4 Wind Power Generation

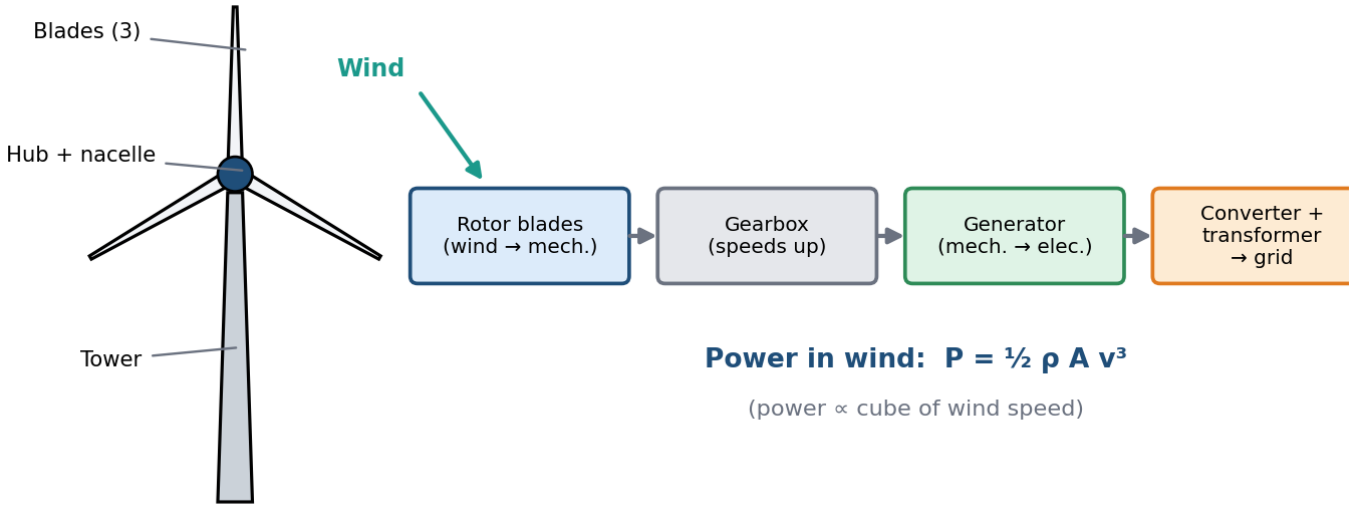


Fig 5 Wind turbine and energy-conversion chain

Moving air turns the turbine blades; the slow rotor speed is increased by a **gearbox** to drive the **generator**. A yaw system turns the nacelle to face the wind. Typical cut-in speed ≈ 3–4 m/s; best sites have steady wind > 6 m/s.

- **Merits:** free and clean fuel, land below can still be farmed, quick to install.
- **Demerits:** wind is intermittent, noise, bird hazard, needs windy sites, high initial cost.

2 ELECTRICITY BILL

2.1 Power Rating of Household Appliances

The **power rating** (in watt, W) is printed on every appliance. Typical (approximate) values:

Appliance	Typical power	Appliance	Typical power
LED bulb	5 – 12 W	Refrigerator	150 – 250 W
Tube light (LED)	20 – 40 W	Washing machine	400 – 600 W
Ceiling fan	60 – 80 W	Electric iron	750 – 1000 W
LED TV	60 – 150 W	Water heater (geyser)	1500 – 3000 W
Desktop PC (with monitor)	150 – 300 W	Microwave oven	1000 – 1400 W
Laptop	40 – 65 W	Air conditioner 1.5 ton	1200 – 1800 W
Printer (laser, printing)	300 – 500 W	Wi-Fi router	5 – 15 W

2.2 The “Unit” of Electrical Energy

DEFINITION
1 unit = 1 kilowatt-hour (kWh) = the energy used by a 1000 W (1 kW) load running for 1 hour. (1 kWh = 3.6×10^6 joule.)

Energy (kWh) = Power (W) × Time (h) / 1000



Fig 6 Steps to calculate an electricity bill

2.3 Two-Part Tariff

The consumer pays two parts: **(1) Fixed (demand) charge** – depends on connected load/sanctioned load in kW, charged per month whether or not energy is used; **(2) Energy charge** – depends on units (kWh) consumed, charged at a rate per unit (often in slabs).

Bill = Fixed charge + (Units consumed × Rate per unit) [+ taxes/duty]

SOLVED EXAMPLE
Q1. Monthly (30-day) bill for a house with: 6 LED lamps (10 W, 5 h/day), 3 fans (75 W, 8 h/day), 1 TV (100 W, 4 h/day), 1 refrigerator (200 W, runs 8 h/day), 1 laptop (60 W, 5 h/day). *Assume* fixed charge ₹100/month, energy charge ₹6/unit.
Lamps: 6×10×5/1000 = 0.30 kWh | Fans: 3×75×8/1000 = 1.80 | TV: 100×4/1000 = 0.40 | Fridge: 200×8/1000 = 1.60 | Laptop: 60×5/1000 = 0.30
Daily = **4.40 kWh** → monthly = 4.4 × 30 = **132 units** → energy charge = 132 × 6 = ₹792 → **Bill = 792 + 100 = ₹892**.
Q2. A 1.5 kW air conditioner runs 8 h/day for 30 days: 1.5 × 8 × 30 = **360 units** (≈ ₹2160 at ₹6/unit).

NOTE
Tariff rates differ from state to state and change often — use the rates given in the question or on your own bill. Saving tip: use 5-star appliances and LED lamps, switch off standby devices.

3 EQUIPMENT SAFETY MEASURES

3.1 Fuse

DEFINITION
A **fuse** is a protective device with a thin wire (element) of low-melting alloy (tin-lead, zinc, silver, copper) connected **in series**. On excessive current, the **heating effect (I^2R)** melts the element and breaks the circuit.

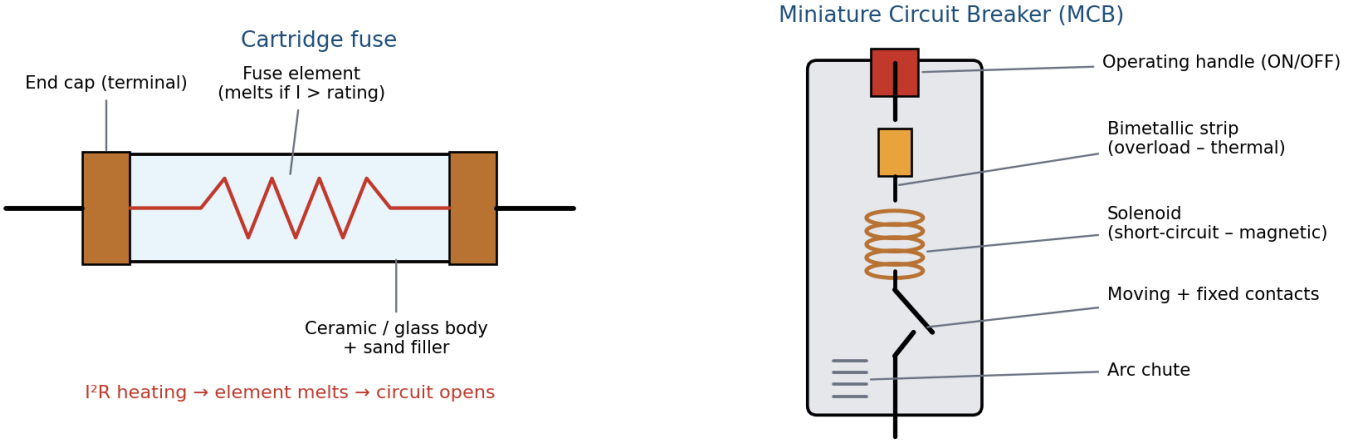


Fig 7 Cartridge fuse (left) and Miniature Circuit Breaker (right)

- **Types:** rewirable (kit-kat) fuse, cartridge (HRC) fuse.
- **Merits:** cheap, simple, no maintenance, very fast on heavy short circuits.
- **Demerits:** must be **replaced** after blowing, slow for small overloads, wrong wire rating can be dangerous, cannot be reset or remotely operated.

3.2 Miniature Circuit Breaker (MCB)

DEFINITION
An **MCB** is an automatic switch that **trips** on overload or short circuit and can be **reset** after the fault is cleared. **Overload:** heated **bimetallic strip** bends and releases the latch (delayed). **Short circuit:** the large current energises a **solenoid** whose plunger trips the contacts instantly. The arc formed is quenched in the **arc chute**.

Type	Trips at (× rated current)	Used for
B	3 – 5 ×	Lighting, resistive loads (homes)
C	5 – 10 ×	Small motors, ACs, mixed loads
D	10 – 20 ×	Transformers, large motors

	Fuse	MCB
Operation	Melts – one-time	Trips – resettable
Response	Slower on small overload	Fast, thermal + magnetic
Cost	Low	Higher
Fault indication	Need to inspect element	Handle position shows
Re-use	Replace element	Switch ON again

3.3 Personal Safety: Electric Shock

DEFINITION

Electric shock occurs when a part of the body completes a path between a live conductor and earth (or between two conductors) so that current flows through it. The **current** (not voltage alone) decides the damage: $I = V / R_{body}$.

Current through body (approx.)	Effect
≈ 1 mA	Threshold of feeling (tingling)
≈ 10 mA	Painful, muscles contract – may be unable to let go
≈ 30 mA	Breathing difficulty
50 – 100 mA	Ventricular fibrillation – may be fatal
> 200 mA	Severe burns, heart stops

Severity depends on: magnitude and duration of current, path through the body (hand-to-hand through chest is worst), skin condition (wet skin resistance ≈ 1 kΩ vs dry ≈ 100 kΩ) and supply type.

3.4 Earthing

DEFINITION

Earthing is the connection of the **non-current-carrying metal parts** of equipment (body, frame) to the general mass of earth through a low-resistance conductor. If insulation fails, the fault current flows safely to earth and the fuse/MCB trips — the body never stays at dangerous voltage.

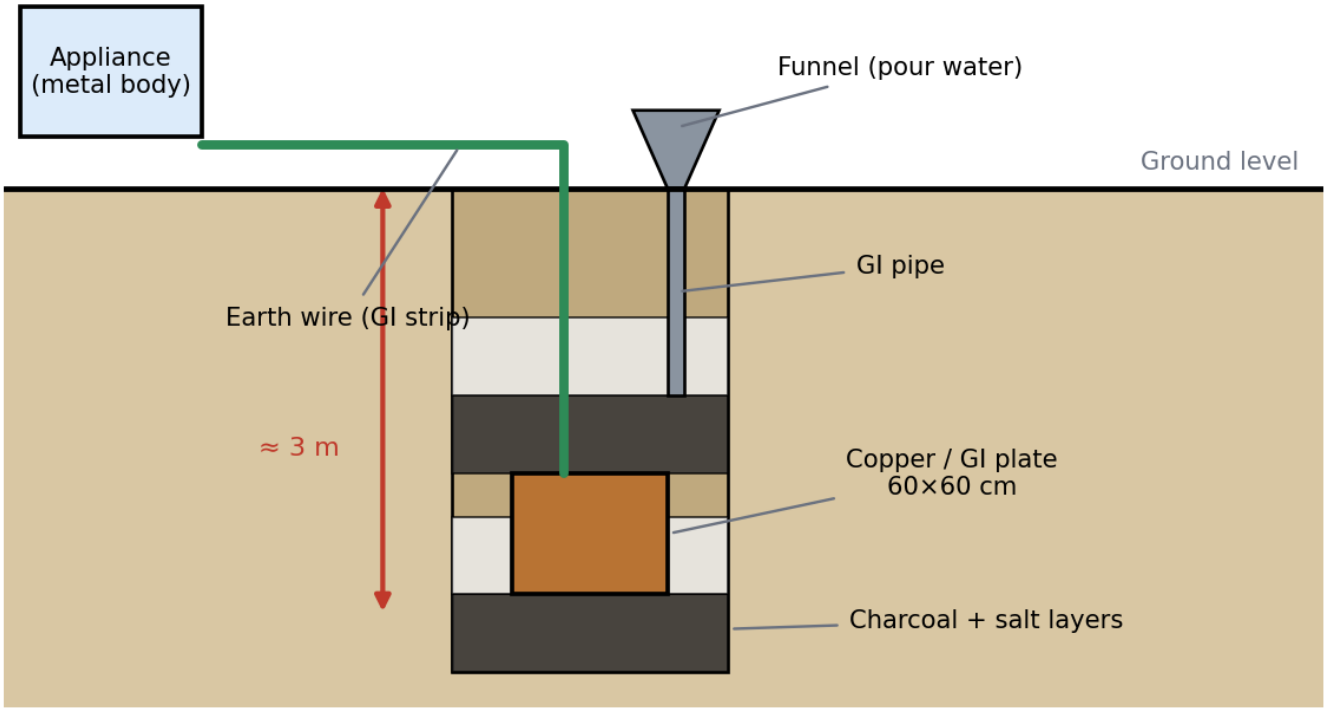


Fig 8 Plate earthing

Type	Details	Typical use
Plate earthing	Copper (60×60×3 mm) or GI (60×60×6 mm) plate buried ≈ 3 m deep with alternate layers of charcoal and salt	Substations, large buildings
Pipe earthing	GI pipe (≈ 38 mm dia, ≈ 2.5 m long) driven vertically; water poured through funnel to keep soil moist	Houses, small installations

Type	Details	Typical use
Rod earthing	Copper/GI rod (12–16 mm dia) driven into ground	Residential, telecom
Strip / wire earthing	Strip buried in trench ≈ 0.5 m deep	Rocky soil, overhead lines

REMEMBER
Earth resistance should be very low (typically < 5 Ω for homes). **Wire colours (India):** Live – Red/Brown, Neutral – Black/Blue, Earth – Green (or Green-Yellow). The earth pin of a 3-pin plug is longer and thicker.

3.5 Safety Precautions to Avoid Shock

- Always **switch off** the main supply before repairing; use a tester to confirm the line is dead.
- Use **proper earthing** and fit **MCB / ELCB (RCCB)** in every installation.
- Never touch switches or appliances with **wet hands** or while standing on a wet floor; use rubber-soled shoes / insulating mats.
- Use **ISI-marked** wires, plugs and appliances; replace damaged or bare wires immediately.
- Never use a fuse wire of the wrong rating; do not bridge a fuse with copper wire.
- Keep water away from sockets; use covered sockets in homes with children; do not overload sockets.
- If someone is stuck to a live wire: **switch off first** or push them away with a dry wooden stick – never touch them directly; then give first aid and call for medical help.

4 QUICK REVISION & EXAM CORNER

Topic	Remember
Hydel	P = 9.81 Q H η kW; Pelton – high head, Kaplan – low head
Nuclear	U-235 fission; moderator slows neutrons; control rods absorb neutrons
Solar / Wind	PV → DC → inverter; wind power ∝ v^3
Unit	1 kWh = 1000 W × 1 h; kWh = W × h / 1000
Bill	Fixed charge + units × rate
Fuse vs MCB	Fuse: I^2R melts, replace; MCB: bimetal + solenoid, resettable
Earthing	Low-resistance path for fault current; plate, pipe, rod, strip

Short questions (2 marks)

- Differentiate conventional and non-conventional energy sources.
- What is the function of the penstock and surge tank?
- What is the role of the moderator and control rods?
- Define 1 unit of electrical energy.
- What is a two-part tariff?
- Write any two demerits of a fuse.
- Why is earthing necessary?

Long questions (5–10 marks)

- Draw the layout of a hydroelectric power plant and explain its working.
- Explain the working of a nuclear power plant with a block diagram.
- Explain solar PV and wind power generation with neat diagrams.

- Explain the working of a fuse and MCB; compare them.
- What is earthing? Explain plate earthing with a neat sketch.
- Calculate the monthly bill for a given load list using a two-part tariff.

Practice numericals (with answers)

No.	Problem	Answer
1	A 2 kW heater runs 3 h daily. Find units used in 30 days.	2×3×30 = 180 units
2	A 40 W tube light and 75 W fan run 10 h daily. Units in 30 days?	(40+75)×10×30/1000 = 34.5 units
3	Fixed charge ₹120, rate ₹5/unit, consumption 200 units. Bill?	120 + 1000 = ₹1120
4	Hydel plant: Q = 20 m^3/s, H = 50 m, η = 80 %. Output power?	9.81×20×50×0.8 = 7848 kW ≈ 7.85 MW